do about it once you had it." Suppressing his deep misgivings about the ethical issues, Oppenheimer, together with the other members of his committee, closed ranks with Teller, Wheeler, and the super's proponents, and the project moved forward at an accelerated pace to construct and test the bomb. It worked as predicted by the combined calculations of Wheeler's team and a cooperating team at Los Alamos.

Wheeler's team's extensive design calculations were ultimately written up as the secret *Project Matterhorn Division B Report 31* or PMB-31. "I'm told," says Wheeler, "that for at least ten years PMB-31 was the bible for design of thermonuclear devices" (hydrogen bombs).

In 1949–50, while America was in a state of panic, and Oppenheimer, Teller, and others were debating whether America should mount a crash program to develop the super, the Soviet Union was already in the midst of a crash superbomb project of its own.

In spring 1948, fifteen months before the first Soviet atomic bomb test, Zel'dovich and his team at the Installation had carried out theoretical calculations on a superbomb design similar to the Americans' "prescription for failure."[3] In June 1948, a second superbomb team was established in Moscow under the leadership of Igor Tamm, one of the most eminent of Soviet theoretical physicists. Its members were Vitaly Ginzburg (of whom we shall hear much in Chapters 8 and 10), Andrei Sakharov (who would become a dissident in the 1970s, and then a hero and Soviet saint in the late 1980s and 1990s), Semyon Belen'ky, and Yuri Romanov. Tamm's team was charged with the task of checking and refining the Zel'dovich team's design calculations.

The Tamm team's attitude toward this task is epitomized by a statement of Belen'ky's at the time: "Our job is to lick Zel'dovich's anus."
Zel'dovich, with his paradoxical combination of a forceful, demanding personality and extreme political timidity, was *not* among the most popular of Soviet physicists. But he *was* among the most brilliant. Landau, who as a leader of a small subsidiary design team occasionally received orders from Zel'dovich's team to analyze this, that, or another facet of the bomb design, sometimes referred to him behind his back as "that bitch, Zel'dovich."
Zel'dovich, by contrast, revered Landau as a great judge of the correctness